

Data Mining Course Project Proposal

Interpretability in "Black Box" Models (XAI)

Prepared scope	Option A: German Credit + 5 papers (2020-2026) Option B: Default of Credit Card Clients + 5 papers (2020-2025)
Note	Because the final dataset has not yet been chosen, this pack contains two complete alternatives. A final recommendation is provided in the Executive Summary.

Objective

This project is fully aligned with the course objective because it selects a benchmark dataset, reviews multiple recent research papers that use the same dataset, analyzes their data mining methodologies in a structured way, and proposes a novel data mining approach based on the identified limitations in the literature. In addition to predictive modeling, the study extends the literature by introducing an explainable AI framework using SHAP and LIME together with stability and faithfulness evaluation. Therefore, the project does not only summarize previous studies, but also proposes an improved and alternative methodology grounded in benchmark-based comparative research.

Executive Summary and Final Recommendation

Both datasets meet the assignment requirements. However, at this stage, Default of Credit Card Clients appears to be the more suitable option for the final project.

This dataset offers a larger sample size, richer repayment and payment-history features, and a broader recent literature base. These strengths make it a strong candidate for developing and evaluating credit scoring models. It is also appropriate for explainability techniques such as SHAP and LIME.

On the other hand, German Credit remains a valuable alternative, especially because it is easier to explain and interpret in an XAI-focused classroom setting. For this reason, while the final decision may depend on the direction of the project, Default of Credit Card Clients currently stands out as the more promising dataset.

Decision factor	Option A - German Credit	Option B - Default of Credit Card Clients
Interpretability friendliness	Very high - smaller feature space and easier borrower narratives	High - richer but more complex payment-history feature set
Sample size / empirical strength	Modest (1,000 rows)	Strong (30,000 rows)
Recent literature depth	Good	Very strong
Best fit for an XAI-heavy course proposal	Excellent if the priority is clear explanations	Excellent if the priority is both strong benchmarking and XAI evaluation
My recommendation	Safer for a small/medium implementation	Best overall final choice if you want the strongest academic case

Option A - Statlog (German Credit Data)

Step 1 - Dataset Selection

This dataset satisfies all Step 1 requirements: it is public, structured, suitable for classification, and heavily reused in the credit-scoring literature. It is especially suitable for the selected XAI topic because model decisions can be traced back to human-readable borrower attributes such as current account status, loan duration, savings, employment, and credit history.

Dataset attribute	Details
Repository	UCI Machine Learning Repository
Task type	Binary classification (good vs bad credit risk)
Instances	1,000
Features	20 predictors (mixed categorical and numerical)
Data format	Structured tabular data
Why it fits the XAI topic	Small, clean, mixed-type credit data makes local and global explanations easy to inspect and discuss in a course project.

Direct dataset link: <https://archive.ics.uci.edu/ml/datasets/Statlog%2B%28German%2BCredit%2BData%29>

Step 2 - Research Paper Collection (2020-2026)

The following five papers all use the same benchmark dataset and satisfy the requirement to provide direct article links.

#	Paper	Year	Venue	Direct link
1	A study on credit scoring modeling with different feature selection and machine learning approaches	2020	Technology in Society	Open article
2	Advancing credit scoring models: integrating explainable AI with strong predictive performance	2024	EAI ECIT Proceedings	Open article
3	Explainable AI For Credit Risk Assessment: Integrating Machine Learning With Business Analytics	2025	IOSR Journal of Economics and Finance	Open article
4	FastTree-Guided Genetic Algorithm for Credit Scoring Feature Selection	2025	Computers (MDPI)	Open article
5	Predictive Modelling of Credit Default Risk Using Machine Learning and Ensemble Techniques	2026	Mathematical and Computational Applications	Open article

Step 3 - Detailed Paper Analysis

Paper 1. A study on credit scoring modeling with different feature selection and machine learning approaches (2020)

Direct link: <https://doi.org/10.1016/j.techsoc.2020.101413>

Problem Definition: Binary credit-scoring classification on the public German credit dataset.

Data Processing: Standard preprocessing was performed before modelling; the paper explicitly evaluates the cleaned German credit dataset with a 66%-34% split and 10-fold cross-validation.

Data Reduction: Feature selection with Chi-Square, Information Gain, and Gain Ratio.

Data Transformation: No separate dimensionality reduction is reported. The accessible text does not fully detail encoding decisions; feature selection is the main reduction step.

Algorithms / Methods: Bayesian classifier, Naive Bayes, SVM-RBF, Decision Tree (C5.0), Random Forest.

Hyperparameters: Exact tuned hyperparameters are not reported.

Evaluation Metrics: Accuracy, F-measure, false-positive rate, false-negative rate, and training time.

Experimental Results: Random Forest + Chi-Square is reported as the best-performing combination; it achieved the strongest overall balance of accuracy/F-measure with low FP and FN rates, although it required more training time.

Paper 2. Advancing credit scoring models: integrating explainable AI with strong predictive performance (2024)

Direct link: <https://doi.org/10.4108/eai.15-3-2024.2346182>

Problem Definition: Improve credit-score prediction while preserving interpretability using a hybrid explainable framework on German Credit.

Data Processing: German Credit with 1,000 records and 20 features; 13 categorical variables are one-hot encoded; 60/40 train-test split.

Data Reduction: No classical dimensionality reduction. The explanatory component uses counterfactual sample generation inside a PermuteAttack framework.

Data Transformation: One-hot encoding of categorical variables; AUC used as the primary evaluation criterion because of class imbalance.

Algorithms / Methods: CatBoost-based hybrid model inside PermuteAttack, plus SHAP and GAN-generated exemplar feedback; compared against Random Forest, Decision Tree, Logistic Regression, LightGBM, AdaBoost, and XGBoost.

Hyperparameters: CatBoost best parameters: depth=15, iterations=1000. Genetic algorithm: population=30, mating parents=10, iterations=100, mutations=1.

Evaluation Metrics: AUC (primary), accuracy, F-score, precision, recall, specificity, false-positive rate.

Experimental Results: CatBoost achieved the best AUC (0.80), outperforming Random Forest (0.77), XGBoost (0.77), Logistic Regression (0.75), AdaBoost (0.72), and Decision Tree (0.59). In the final framework, CatBoost also improved accuracy (0.7825), F-score (0.5700), precision (0.7123), and specificity (0.9255) over Random Forest.

Paper 3. Explainable AI For Credit Risk Assessment: Integrating Machine Learning With Business Analytics (2025)

Direct link: <https://www.iosrjournals.org/iosr-jef/papers/Vol16-Issue5/Ser-1/H1605016372.pdf>

Problem Definition: Transparent credit-risk classification on German Credit with both predictive and business-analytics interpretation goals.

Data Processing: German Credit data (1,000 borrower records). Categorical fields were encoded, continuous variables standardized, dummy variables created where needed, inconsistencies resolved, missing values handled, variables normalized, and train/test subsets prepared. Models were also evaluated on a balanced dataset using 10-fold cross-validation.

Data Reduction: No separate dimensionality reduction method is reported.

Data Transformation: Encoding, standardization, normalization, dummy-variable generation.

Algorithms / Methods: Logistic Regression, Random Forest, XGBoost, and MLP/Neural Network, followed by SHAP and LIME interpretation.

Hyperparameters: Hyperparameters are not reported.

Evaluation Metrics: Accuracy, precision, recall, F1-score, AUC-ROC, mean CV accuracy, standard deviation.

Experimental Results: Logistic Regression produced the highest accuracy (77.33%), precision (76.26%), recall (77.33%), and F1-score (75.98%). Random Forest slightly led on AUC-ROC (0.7755 vs 0.7743 for Logistic Regression) and achieved the highest mean CV accuracy (76.6%, sd=1.91%). The SHAP summary identified amount, age, and duration as the strongest global drivers.

Paper 4. FastTree-Guided Genetic Algorithm for Credit Scoring Feature Selection (2025)

Direct link: <https://doi.org/10.3390/computers14120566>

Problem Definition: Feature-selection optimisation for credit scoring across benchmark datasets, including German Credit.

Data Processing: Cross-dataset benchmark study; German dataset analysed with multiple feature-selection pipelines and 10-fold cross-validation.

Data Reduction: Chi2, mutual information, RFE, L1-LogReg, Random-Forest selection, and the proposed FT-GA feature-selection method.

Data Transformation: Dataset-specific preprocessing; the accessible text emphasises feature-subset optimisation rather than additional transformation.

Algorithms / Methods: FT-GA feature selector paired with FastTree; baseline downstream models include Logistic Regression, Naive Bayes, Random Forest, and Neural Network.

Hyperparameters: German GA configuration: population size=50, crossover rate=0.8, mutation rate=0.05, tournament size=3.

Evaluation Metrics: Accuracy, precision, recall, F1-score, AUC, KS statistic, 10-fold CV mean accuracy and mean AUC.

Experimental Results: For the German dataset, FT-GA + FastTree achieved accuracy=0.8336, precision=0.8642, recall=0.9429, F1=0.9018, AUC=0.8306 in the single evaluation table. In 10-fold CV, FT-GA reached mean accuracy=0.8194 and mean AUC=0.8447, outperforming the listed German baselines.

Paper 5. Predictive Modelling of Credit Default Risk Using Machine Learning and Ensemble Techniques (2026)

Direct link: <https://doi.org/10.3390/mca31020045>

Problem Definition: Balance predictive accuracy and interpretability in German-credit default modelling using ensembles and XAI.

Data Processing: German Credit with 1,000 applications and a 30% default rate. Categorical variables encoded, numeric features scaled, class imbalance handled with SMOTE, and thresholds optimized using Youden's J.

Data Reduction: No dimensionality reduction; model comparison plus threshold optimisation are the main methodology steps.

Data Transformation: Feature encoding, scaling, class-weighting/imbalance handling, threshold optimisation.

Algorithms / Methods: Logistic Regression, Random Forest, XGBoost, MLP, and a Stacked Ensemble with logistic-regression meta learner; SHAP for global/local interpretability.

Hyperparameters: Logistic Regression: penalty=l2, C=1.0. Random Forest: n_estimators=100, max_depth=10. XGBoost: learning_rate=0.1, max_depth=6, subsample=0.9. MLP search includes hidden-layer-size tuning.

Evaluation Metrics: Accuracy, precision, recall, F1-score, AUC, confusion matrices, Friedman test ranks.

Experimental Results: The Stacked Ensemble achieved the best base-model AUC (0.761) with precision=0.783, recall=0.806, F1=0.794. Random Forest ranked second (AUC=0.749). SHAP showed Current Account status, Loan Duration, and Savings Account as the dominant predictors. Cost-sensitive threshold optimisation reduced false positives from 39 to 16.

Step 4 - Comparative Table

Compact comparison of the five reviewed papers for this dataset option.

Paper	Year	Main method	Data processing	Data reduction	Transformation	Key hyperparameters	Best reported result
Trivedi	2020	Bayesian classifier, Naive Bayes, SVM-RBF, Decision Tree (C5.0), RandomForest.	Standard preprocessing was performed before modelling; the paper explicitly evaluates the cleaned German credit dataset with a 66%-34% split and 10-fold cross-validation.	Feature selection with Chi-Square, Information Gain, and Gain Ratio.	No separate dimensionality reduction is reported. The accessible text does not fully detail encoding decisions; feature selection is the main reduction step.	Exact tuned hyperparameters are not reported.	Random Forest + Chi-Square is reported as the best-performing combination; it achieved the strongest overall balance of accuracy/F-measure with low FP and FN rates, although it required more training time.
Zhu et al.	2024	CatBoost-based hybrid model inside PermuteAttack, plus SHAP and GAN-generated exemplar feedback; compared against Random Forest, Decision Tree, Logistic Regression, LightGBM, AdaBoost, and XGBoost.	German Credit with 1,000 records and 20 features; 13 categorical variables are one-hot encoded; 60/40 train-test split.	No classical dimensionality reduction. The explanatory component uses counterfactual sample generation inside a PermuteAttack framework.	One-hot encoding of categorical variables; AUC used as the primary evaluation criterion because of class imbalance.	CatBoost best parameters: depth=15, iterations=1000. Genetic algorithm: population=30, mating parents=10, iterations=100, mutations=1.	CatBoost achieved the best AUC (0.80), outperforming Random Forest (0.77), XGBoost (0.77), Logistic Regression (0.75), AdaBoost (0.72), and Decision Tree (0.59). In the final framework, CatBoost also improved accuracy (0.7825), F-score (0.5700), precision (0.7123), and specificity (0.9255) over Random Forest.

Oyeyemi et al.	2025	Logistic Regression, RandomForest, XGBoost, and MLP/Neural Network, followed by SHAP and LIME interpretation.	German Credit data (1,000 borrower records). Categorical fields were encoded, continuous variables standardized, dummy variables created where needed, inconsistencies resolved, missing values handled, variables normalized, and train/test subsets prepared. Models were also evaluated on a balanced dataset using 10-fold cross-validation.	No separate dimensionality reduction method is reported.	Encoding, standardization, normalization, dummy-variable generation.	Hyperparameters are not reported.	Logistic Regression produced the highest accuracy (77.33%), precision (76.26%), recall (77.33%), and F1-score (75.98%). Random Forest slightly led on AUC-ROC (0.7755 vs 0.7743 for Logistic Regression) and achieved the highest mean CV accuracy (76.6%, sd=1.91%). The SHAP summary identified amount, age, and duration as the strongest global drivers.
Bahlool et al.	2025	FT-GA feature selector paired with FastTree; baseline downstream models include Logistic Regression, Naive Bayes, RandomForest, and Neural Network.	Cross-dataset benchmark study; German dataset analysed with multiple feature-selection pipelines and 10-fold cross-validation.	Chi2, mutual information, RFE, L1-LogReg, RandomForest selection, and the proposed FT-GA feature-selection method.	Dataset-specific preprocessing; the accessible text emphasises feature-subset optimisation rather than additional transformation.	German GA configuration: population size=50, crossover rate=0.8, mutation rate=0.05, tournament size=3.	For the German dataset, FT-GA + FastTree achieved accuracy=0.8336, precision=0.8642, recall=0.9429, F1=0.9018, AUC=0.8306 in the single evaluation table. In 10-fold CV, FT-GA reached mean accuracy=0.8194 and mean AUC=0.8447, outperforming the listed German baselines.

	Mathibela & Maposa	Logistic Regression, Random Forest, XGBoost, MLP, and a Stacked Ensemble with logistic-regression meta learner; SHAP for global/local interpretability	German Credit with 1,000 applications and a 30% default rate. Categorical variables encoded, numeric features scaled, class imbalance handled with SMOTE, and thresholds optimized using Youden's J.	No dimensionality reduction; model comparison plus threshold optimisation are the main methodology steps.	Feature encoding, scaling, class-weighting/imbalance handling, threshold optimisation.	Logistic Regression: penalty=l2, C=1.0. Random Forest: n_estimators=100, max_depth=10. XGBoost: learning_rate=0.1, max_depth=6, subsample=0.9. MLP search includes hidden-layer-size tuning.	The Stacked Ensemble achieved the best base-model AUC (0.761) with precision=0.783, recall=0.806, F1=0.794. Random Forest ranked second (AUC=0.749). SHAP showed Current Account status, Loan Duration, and Savings Account as the dominant predictors. Cost-sensitive threshold optimisation reduced false positives from 39 to 16.
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Step 5 - Critical Analysis

1. Which methods perform best and why?

Across the five papers, tree-based ensembles and hybrid methods dominate. Random Forest + Chi-Square is best in Trivedi (2020), CatBoost leads in Zhu et al. (2024), FT-GA + FastTree is strongest in Bahlool et al. (2025), and the stacked ensemble leads in Mathibela & Maposa (2026). The pattern is consistent: German Credit is small, mixed-type, and nonlinear enough that ensemble learners outperform purely linear baselines while still keeping enough structure for meaningful explanations.

2. Which preprocessing techniques are most common?

Encoding categorical variables is the most universal step. Standardization/scaling appears whenever neural or distance-sensitive models are used, while class-imbalance handling appears more frequently in the recent papers (SMOTE, threshold optimisation, balanced evaluation). Feature selection is also common, especially in older comparative work and in the FT-GA paper.

3. What limitations exist in current research?

Three recurring limitations stand out. First, the dataset itself is small (1,000 rows), which restricts generalisability. Second, many papers show feature-importance plots but do not formally evaluate explanation quality or stability. Third, several papers optimize prediction accuracy but do not integrate calibration, cost-sensitive thresholds, and faithfulness testing into one coherent methodology.

4. What patterns exist across papers?

There is a clear shift over time. The 2020 paper emphasizes classical model selection and feature selection. The 2024-2026 papers increasingly combine stronger ensembles with XAI, class-imbalance handling, and practical decision framing. This makes German Credit a very good benchmark for proposing an XAI-aware methodology that is still compact enough for a student project.

Step 6 - Proposed Methodology

1. Proposed Pipeline

- Data audit and cleaning: verify category consistency, missing-value treatment, and outlier checks.
- Hybrid preprocessing: one-hot encode nominal variables, preserve ordinal order where appropriate, standardize numeric variables, and compare class_weight against SMOTE-ENN.
- Feature engineering: debt burden proxies, installment-to-loan ratio, and interaction features between current-account status, savings, and duration.
- Model training: Logistic Regression baseline, Random Forest, XGBoost, and LightGBM, followed by a calibrated stacking ensemble.
- Nested cross-validation and Bayesian/random hyperparameter search to avoid optimistic bias.
- Explainability layer: global SHAP, local SHAP waterfall plots, LIME explanations for borderline borrowers, plus explanation-stability tests.
- Faithfulness evaluation: remove or perturb top-ranked features and measure probability shift.
- Final evaluation: Accuracy, Precision, Recall, F1, ROC-AUC, calibration error, and explanation stability.

2. Novelty / Difference from the Literature

- Most reviewed German-credit papers either optimize prediction or add one explainability tool. The proposed method combines strong predictive modelling, calibration, and a formal explanation-evaluation stage in one pipeline.
- Unlike papers that only show SHAP/LIME plots, this proposal explicitly measures explanation stability and faithfulness, which is a stronger XAI contribution for a project proposal.
- The project also adds threshold optimisation and calibration, which are important in credit decisions but are not consistently integrated across the reviewed papers.

3. Expected Contribution

Expected outcome: a credit-risk model that remains competitive on AUC/F1 while producing reproducible global and local explanations that can be defended in a high-stakes financial setting.

Option B - Default of Credit Card Clients (Taiwan)

This section is written to match the exact structure requested in the course instruction sheet (Step 1 to Step 6).

Step 1 - Dataset Selection

This dataset also fully satisfies Step 1: it is public, structured, widely benchmarked, and suitable for data mining classification. Compared with German Credit, it provides substantially more records and a richer temporal payment-history structure, which is especially useful if the final project needs stronger empirical performance or a more convincing modern benchmark.

Dataset attribute	Details
Repository	UCI Machine Learning Repository
Task type	Binary classification (default next month vs non-default)
Instances	30,000
Features	23 predictors (plus target)
Data format	Structured tabular data
Why it fits the XAI topic	Larger scale, richer payment-history variables, and stronger recent literature make it ideal for comparing black-box performance with SHAP/LIME explanations.

Direct dataset link: <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

Step 2 - Research Paper Collection (2020-2026)

The following five papers all use the same benchmark dataset and satisfy the requirement to provide direct article links.

#	Paper	Year	Venue	Direct link
1	An Investigation of Credit Card Default Prediction in the Imbalanced Datasets	2020	IEEE Access	Open article
2	Research on Credit Card Default Prediction Based on k-Means SMOTE and BP Neural Network	2021	Complexity	Open article
3	Toward interpretable credit scoring: integrating explainable artificial intelligence with deep learning for credit card default prediction	2023	Neural Computing and Applications	Open article
4	Credit Card Default Prediction: An Empirical Analysis on Predictive Performance Using Statistical and Machine Learning Methods	2025	Journal of Risk and Financial Management	Open article
5	SHAP Stability in Credit Risk Management: A Case Study in Credit Card Default Model	2025	Risks	Open article

Step 3 - Detailed Paper Analysis

Paper 1. An Investigation of Credit Card Default Prediction in the Imbalanced Datasets (2020)

Direct link: <https://doi.org/10.1109/ACCESS.2020.3033784>

Problem Definition: Default prediction on imbalanced credit-card data, including the Taiwan default dataset.

Data Processing: Min-max normalization is applied to numeric features; training/test split validation is used; imbalance is handled with multiple under- and over-sampling techniques.

Data Reduction: No dimensionality reduction is reported.

Data Transformation: Normalization plus resampling transformations (Random Over/Under, ADASYN, SMOTE, Borderline-SMOTE, SMOTE-Tomek, K-means SMOTE, Near Miss, Cluster Centroids).

Algorithms / Methods: GBDT plus comparisons with Random Forest, Bagging, KNN, Logistic Model Tree, AdaBoost, and Stacking under multiple imbalance-handling strategies.

Hyperparameters: Learning rate = 0.2 and number of constructed trees = 200.

Evaluation Metrics: Accuracy, precision, recall, F-measure, ROC, G-mean, plus One-way ANOVA significance testing.

Experimental Results: On the imbalanced Taiwan dataset the baseline accuracy is reported as 66.9%; proposed rebalanced models improve this to about 89%. GBDT with K-means SMOTE is highlighted as the strongest configuration for Taiwan, with reported accuracy around 88.7%.

Paper 2. Research on Credit Card Default Prediction Based on k-Means SMOTE and BP Neural Network (2021)

Direct link: <https://doi.org/10.1155/2021/6618841>

Problem Definition: Improve default prediction on imbalanced credit-card data through oversampling and neural-network optimisation.

Data Processing: The paper addresses class imbalance directly by applying k-means SMOTE before modelling.

Data Reduction: No classical dimensionality reduction is reported.

Data Transformation: Data distribution is altered with k-means SMOTE; feature importance from Random Forest is injected into BP network initial weights.

Algorithms / Methods: Proposed model = k-means SMOTE + BP neural network; compared against KNN, logistic model, SVM, Random Forest, and tree-based baseline models.

Hyperparameters: Default values.

Evaluation Metrics: AUC and classification performance comparisons across six models.

Experimental Results: The proposed method increases AUC from 0.765 to 0.929. The authors conclude that BP neural network provides the best overall predictive effect among the tested models once imbalance handling is applied.

Paper 3. Toward interpretable credit scoring: integrating explainable artificial intelligence with deep learning for credit card default prediction (2024)

Direct link: <https://doi.org/10.1007/s00521-023-09232-2>

Problem Definition: Credit-card default prediction with a deliberate balance between deep-learning accuracy and interpretability.

Data Processing: Uses the UCI Default of Credit Card Clients dataset and evaluates a real-world predictive framework on that benchmark.

Data Reduction: No separate dimensionality reduction is stated in the visible preview.

Data Transformation: The preview confirms an XAI-enhanced deep-learning pipeline; specific preprocessing and transformations beyond the abstract are not fully visible in the free preview.

Algorithms / Methods: Deep learning combined with explainable AI (XAI); the paper explicitly frames the method as interpretable credit scoring and references SHAP as a core keyword.

Hyperparameters: Not clearly available in the accessible preview.

Evaluation Metrics: Accuracy, sensitivity, specificity, and comparison to state-of-the-art baselines.

Experimental Results: The reported model achieves accuracy=0.8350, sensitivity=0.8823, and specificity=0.9879. The most important features are payment delays and outstanding bill amounts, supporting strong financial interpretability.

Paper 4. Credit Card Default Prediction: An Empirical Analysis on Predictive Performance Using Statistical and Machine Learning Methods (2025)

Direct link: <https://doi.org/10.3390/jrfm18010023>

Problem Definition: Compare traditional statistical models and machine-learning models for Taiwan credit-card default prediction.

Data Processing: Dataset contains 30,000 records (6636 defaults, 23,364 non-defaults), no duplicates or missing values, and is imbalanced. Feature scaling is performed using StandardScaler.

Data Reduction: No explicit dimensionality reduction; LDA is used as a classifier rather than a separate reduction stage.

Data Transformation: StandardScaler normalization on input variables.

Algorithms / Methods: LDA, Logistic Regression, SVM, XGBoost, Random Forest, and Deep Neural Network.

Hyperparameters: DNN best setup: 4 hidden layers, ReLU activations, 25 epochs. Other models are implemented in Python; exact parameter grids for every classical model are not fully summarized in the visible excerpt.

Evaluation Metrics: Accuracy, sensitivity/recall, specificity, precision, F1, G-mean, ROC, and AUC; 10-fold CV.

Experimental Results: DNN reached the best cross-validated accuracy (81.80%) and best F1 (0.4820) and G-mean (0.6027). DNN and XGBoost tied for the highest AUC (0.77). The paper also uses feature importance to open the DNN “black box”, finding BillSum and age as the strongest features.

Paper 5. SHAP Stability in Credit Risk Management: A Case Study in Credit Card Default Model (2025)

Direct link: <https://doi.org/10.3390/risks13120238>

Problem Definition: Evaluate how stable SHAP explanations remain when the same credit-default task is trained repeatedly with different random seeds.

Data Processing: Uses the 30,000-record UCI Taiwan dataset. Education categories are merged, exploratory cleaning is performed, one-hot encoding is applied, and no resampling is used because the default rate (22.22%) is considered manageable. Final modelling matrix has 79 predictors after encoding.

Data Reduction: No dimensionality reduction; the study focuses on explainability stability instead.

Data Transformation: Exploratory cleaning, variable selection, variable transformation, one-hot encoding of categorical variables.

Algorithms / Methods: 100 XGBoost models with identical hyperparameters but different random seeds, followed by global SHAP ranking analysis and Kendall’s W stability analysis.

Hyperparameters: Learning rate = 0.1, maximum tree depth = 5, subsample = 0.8, colsample_bytree = 0.8, boosting rounds = 100.

Evaluation Metrics: Accuracy, F1-score, KS statistic, AUROC, confusion-matrix medians and confidence intervals, Kendall’s W for ranking stability.

Experimental Results: Median performance includes accuracy=79.53% and AUROC=77.73%. Overall SHAP concordance is extremely high (Kendall’s W=0.98 across all features). The top five features also remain highly stable (W=0.93), with limit_bal ranking first in 96 out of 100 models.

Step 4 - Comparative Table

Compact comparison of the five reviewed papers for this dataset option.

Paper	Year	Main Method	Data Processing	Data Reduction	Transformation	Key Hyperparameters	Best Reported Result
Alam et al.	2020	GBDT plus comparisons with RandomForest, Bagging, KNN, Logistic Model Tree, AdaBoost, and Stacking under multiple imbalance-handling strategies.	Min-max normalization is applied to numeric features; training/test split validation is used; imbalance is handled with multiple under- and over-sampling techniques.	No dimensionality reduction is reported.	Normalization plus resampling transformations (Random Over/Under, ADASYN, SMOTE, Borderline-SMOTE, SMOTE-Tomek, K-means SMOTE, Near Miss, Cluster Centroids).	Learning rate = 0.2 and number of constructed trees = 200.	On the imbalanced Taiwan dataset the baseline accuracy is reported as 66.9%; proposed rebalanced models improve this to about 89%. GBDT with K-means SMOTE is highlighted as the strongest configuration for Taiwan, with reported accuracy around 88.7%.
Chen & Zhang	2021	Proposed model = k-means SMOTE + BP neural network; compared against KNN, logistic model, SVM, RandomForest, and tree-based baseline models.	The paper addresses class imbalance directly by applying k-means SMOTE before modelling.	No classical dimensionality reduction is reported.	Data distribution is altered with k-means SMOTE; feature importance from Random Forest is injected into BP network initial weights.	Default values.	The proposed method increases AUC from 0.765 to 0.929. The authors conclude that BP neural network provides the best overall predictive effect among the tested models once imbalance handling is applied.
Talaat et al.	2023	Deep learning combined with explainable AI (XAI); the paper explicitly frames the method as interpretable credit scoring and references SHAP as a core keyword.	Uses the UCI Default of Credit Card Clients dataset and evaluates a real-world predictive framework on that benchmark.	No separate dimensionality reduction is stated in the visible preview.	The preview confirms an XAI-enhanced deep-learning pipeline; specific preprocessing and transformations beyond the abstract are not fully visible in the free preview.	Hyperparameters are not reported.	The reported model achieves accuracy=0.8350, sensitivity=0.8823, and specificity=0.9879. The most important features are payment delays and outstanding bill amounts, supporting strong financial interpretability.

Bhandary & Ghosh	2025	LDA, Logistic Regression, SVM, XGBoost, RandomForest, and Deep Neural Network.	Dataset contains 30,000 records (6636 defaults, 23,364 non-defaults), no duplicates or missing values, and is imbalanced. Feature scaling is performed using StandardScaler.	No explicit dimensionality reduction; LDA is used as a classifier rather than a separate reduction stage.	StandardScaler normalization on input variables.	DNN best setup: 4 hidden layers, ReLU activations, 25 epochs. Other models are implemented in Python; exact parameter grids for every classical model.	DNN reached the best cross-validated accuracy (81.80%) and best F1 (0.4820) and G-mean (0.6027). DNN and XGBoost tied for the highest AUC (0.77). The paper also uses feature importance to open the DNN “black box”, finding BillSum and age as the strongest features.
Lin & Wan	2025	100 XGBoost models with identical hyperparameters but different randomseeds, followed by global SHAP ranking analysis and Kendall's W stability analysis.	Uses the 30,000-record UCI Taiwan dataset. Education categories are merged, exploratory cleaning is performed, one-hot encoding is applied, and no resampling is used because the default rate (22.22%) is considered manageable. Final modelling matrix has 79 predictors after encoding.	No dimensionality reduction; the study focuses on explainability stability instead.	Exploratory cleaning, variable selection, variable transformation, one-hot encoding of categorical variables.	Learning rate = 0.1, maximum tree depth = 5, subsample = 0.8, colsample_bytree = 0.8, boosting rounds = 100.	Median performance includes accuracy=79.53% and AUROC=77.73%. Overall SHAP concordance is extremely high (Kendall's W=0.98 across all features). The top five features also remain highly stable (W=0.93), with limit_bal ranking first in 96 out of 100 models.

Step 5 - Critical Analysis

1. Which methods perform best and why?

Recent Taiwan-dataset papers show that modern ensemble and deep-learning models usually outperform classical linear models on AUC, F1, or recall. However, the “best” method depends on the goal: Alam et al. (2020) show resampling can dramatically lift recall/accuracy on the minority class, Bhandary & Ghosh (2025) find DNN strongest overall, and Lin & Wang (2025) show that explanation robustness itself becomes an important evaluation dimension beyond raw score maximization.

2. Which preprocessing techniques are most common?

The most common steps are class-imbalance handling, scaling/normalization, categorical cleanup of anomalous codes, and feature engineering around payment behavior. Compared with German Credit, the Taiwan dataset has richer temporal payment variables, so payment-history transformation and aggregate bill/payment features are more natural.

3. What limitations exist in current research?

Most papers still optimize one aspect at a time: imbalance handling, deep learning, or XAI. Few papers jointly evaluate predictive performance, explanation stability, calibration, and operational cost trade-offs. Another limitation is that many studies report one-off SHAP/LIME figures without testing whether explanations remain stable across repeated training runs.

4. What patterns exist across papers?

The literature trend is strong and useful for this proposal: 2020-2021 papers focus on imbalance correction and neural prediction, while 2024-2025 papers move toward interpretable deep learning, black-box opening, and SHAP stability. This gives Option B a stronger modern literature arc and a very solid justification for an XAI-oriented project proposal.

Step 6 - Proposed Methodology

The following proposal is intentionally different from the reviewed papers while staying realistic for a student data-mining project.

1. Proposed Pipeline

- Data audit and category repair: merge anomalous EDUCATION/MARRIAGE codes, inspect negative bill values, and keep domain-justified exceptions with clear documentation.
- Feature engineering: payment-delay aggregates, bill-to-payment ratios, rolling payment intensity, total billed amount, total paid amount, and utilization-style proxies.
- Preprocessing alternatives: compare no-resampling, class weighting, SMOTE, and SMOTE-ENN inside nested cross-validation.
- Model training: Logistic Regression baseline, Random Forest, XGBoost, LightGBM, and a calibrated deep tabular model or stacking ensemble.
- Explainability layer: SHAP for global and local explanations; LIME for borderline cases; SHAP stability analysis across repeated seeds following Lin & Wang (2025).
- Explanation evaluation: Kendall's τ / rank correlation for stability, probability-shift tests for faithfulness, and comparison of explanation patterns between correctly and incorrectly classified defaults.
- Final evaluation: Accuracy, Recall, Precision, F1, ROC-AUC, PR-AUC, calibration, confusion costs, and explanation stability.

2. Novelty / Difference from the Literature

- This proposal directly builds on the strongest gaps found in the recent Taiwan-dataset literature: papers either optimize imbalance handling, or use deep learning, or discuss XAI, but rarely combine prediction, calibration, and formal explanation-quality testing in one framework.
- Adding repeated-seed SHAP stability and local LIME contrastive analysis would create a more rigorous XAI contribution than standard feature-importance plots alone.
- The richer payment-history variables also allow new financially interpretable engineered features that most benchmark papers still underuse.

3. Expected Contribution

Expected outcome: a stronger large-sample benchmark than German Credit, with better empirical depth and a clearer opportunity to compare black-box performance against explanation consistency on a realistic default dataset.

